

Matrix Scroll & SSX360

Insider Rollout Packet & Strategic Briefing

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1. Executive Summary & Dual-Co Structure

This strategic document outlines the product partition, current live state, and federal compliance roadmap for **Matrix Scroll** and **SSX360 (Pegasus Devices)**. To optimize venture scalability and navigate rigorous federal certification lines, the entities operate under a dual operational model sharing a single Ed25519 cryptographic primitive:

- **Company 1: Matrix Scroll (SaaS)**: Focuses purely on software-driven developer hooks, MCP integration modules, and a globally trusted SaaS Hosted Identity Authority. Drives recurring high-margin cloud revenue.
- **Company 2: SSX 360 (Pegasus Devices)**: Focuses on physical cryptographic hardening. Isolates standard Ed25519 primitives into physical secure element silicon, capturing high-security enterprise and federal clients.

2. Current State: What is Done and Live

A. Software Layer (matrixscroll.com)

- **Obsidian Dark Redesign**: Visual upgrades completed across documentation, spec, and verifier pages using Inter/JetBrains Mono fonts, custom geometric brand marks, and optimized grid layouts.
- **PyPI SDK 0.2.6**: Reference Python SDK is live on PyPI, offering pre-commit hooks and offline verifier tools.
- **CI Action & Browser Verifier**: Github Action is operational for PR gating. Client-side browser verifier is live at matrixscroll.com/verify/ without database or backend requirements.
- **Attribution & Contact**: Link to *Ryan James David York* and contact address security@matrixscroll.com are successfully published in all page footers.

B. Hardware Layer (ssx360.com)

- **Partitioning**: All commercial pricing models, account signup structures, and hardware mockups have been deleted from the open-source website. The reference device preview screen is live at ssx360.com.

3. Product & Commercial Frameworks

A. Matrix Scroll SaaS Tiers

Tier	Price	Features	Focus
Freedom	\$0	Local SDK, self-signing, MCP modules, basic local checking hook	Developer Adoption
Basic	\$25/mo	Hosted authority: verified identity certificate for individuals	Individual Verification
Team	\$40/seat	NFC smartcard support, team authorization ledger logs	Team Verification
Enterprise	Custom	On-prem secure CA authority, AMD-SEV attestation checks	Accredited Enclaves

B. Pegasus Hardware Specifications

Device Type	Financials	Description & Form Factor
Pegasus Card (NFC)	COGS: \$2.90 Retail: \$29.00	NFC secure smartcard generating on-chip signature envelopes. Perfect for badge or touch authorization.
Crystal Dongle	COGS: \$13.50 Retail: \$99.00	USB-C key utilizing embedded NXP SE050 modules. Plugs into local nodes for cryptographically attested commits.
IoT Sensor Seal	COGS: \$45.00 Retail: \$299.00	Solderable hardware modules signing data streams directly. Verifies device payloads to prevent spoofing.

4. Federal & Regulatory Alignment

Matrix Scroll and Pegasus map directly into federal cybersecurity directives, addressing CMMC 2.0 control gaps and supply chain requirements:

Mandate	Control / Clause	How Matrix Scroll / Pegasus Satisfies It
EO 14028 §4	Software attestation	DSSE-compatible signed envelopes per commit, per build, per mandate.
OMB M-22-18 / M-23-16	Producer self-attestation	Cryptographic receipt replaces PDF attestation; verifiable by agency offline.
NIST SP 800-218 (SSDF)	PS.2, PW.4, RV.1	Authorship signatures, key custody on SE050, verifiable provenance log.
NIST SP 800-207 (ZTA)	NHI / device identity	Per-agent hardware-bound identity; replaces shared service-account tokens.
CMMC 2.0 L2 / L3	IA.L2-3.5.x, SC.L2-3.13.x	Hardware-isolated private keys, FIPS-validated crypto path (2027), per-action auth.
FIPS 140-3	Module validation	Pegasus module on roadmap for L2 certification; CAVP algorithm validation interim.
CNSA 2.0 / PQC	Post-quantum readiness	Envelope format reserves algorithm field; ML-DSA (Dilithium-2) co-signing planned for 2027.

5. Compliance & Rollout Roadmap

- **Q2 2026 (Live):** Base engine live on PyPI (v0.2.6), MCP server hooks, and web verifier operational.
- **Q4 2026:** Cryptographic Algorithm Validation Program (CAVP) algorithm validation submission.
- **Q4 2026:** Launch Hosted Certificate Authority (CA) architecture.
- **H1 2027:** Pegasus hardware tokens (Cards & Dongles) graduate to General Availability (GA).
- **Q2 2027:** FIPS 140-3 Level 2 module certification target.
- **Q3 2027:** Common Criteria EAL4+ (carry-over from SE050 base) target.
- **H2 2027:** Edge Attestation (AMD-SEV cloud enclaves checking transaction lines) live.
- **2027 Roadmap:** Post-quantum hybrid signatures (ML-DSA / Dilithium-2) co-signing integration.

6. Must-Dos and Action Plan

The following checklist details the immediate actions for the hardware and Digital Rain rollout on ssx360.com:

- [] Refactor local daemon to automatically detect USB/HID FIDO2 keys
- [] Develop local compiler/bindings for NXP SE050 cryptographic signing hooks
- [] Establish L3 attestation schemas including hardware certificate chains
- [] Deploy the final ssx360.com landing page with onboarding setup wizard